Monitoring Urban Change in Informal Settlements Using Street-Level Imagery

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Abstract

Informal settlements house over one billion people globally yet remain inadequately monitored by traditional data sources. This research develops a methodology to quantify built environment change in informal settlements using Google Street View (GSV) imagery and machine learning. We address two questions: To what extent does GSV capture informal areas and their temporal evolution? What can be reliably measured from street-level imagery in continuously transforming environments? Integrating GSV panoramas with Argentina's ReNaBaP registry and semantic segmentation models, we analyze spatial coverage, temporal representation, and physical change indicators. A pilot study in Tres de Febrero, Buenos Aires, demonstrates the feasibility across 14 neighborhoods, documenting measurable changes in building density over time. The project proposes scaling this methodology to all 6,467 registered informal settlements nationwide, generating the first comprehensive street-level longitudinal dataset for Argentina.

Introduction

Informal settlements—often referred to as slums, favelas, or shantytowns—represent one of the most persistent and complex challenges of urbanization in the Global South (Smit, 2021). They are typically characterized by insecure land tenure, inadequate access to basic services and infrastructure, substandard housing materials, and limited formal recognition by local authorities (UN-Habitat, 2016). Despite their heterogeneity, informal settlements share a common feature: they are products of rapid urbanization coupled with social, political, and economic exclusion (Acevedo-De-los-Ríos et al., 2025). Residents build incrementally, adapting to evolving circumstances and constraints, which makes these environments highly dynamic in both spatial and temporal terms.

Governments, international organizations, and local communities have long recognized the need to address the challenges of informal settlements through policies that promote upgrading, regularization, and inclusive urban planning (Huchzermeyer & Karam, 2006). However, the design and implementation of such policies depend heavily on reliable, up-to-date information about where informal settlements are located, how they evolve, and what conditions residents face.

Monitoring and measuring informal settlements have often y relied on a combination methods, including field surveys, aerial photography, and increasingly, satellite imagery (Sartori et al., 2002). Traditional data sources—including censuses, household surveys, and cadastral maps—often fail to capture the full extent of informal urbanization (Thomson et al., 2020). They are costly, infrequent, and tend to omit spatial and temporal details that are critical for understanding intra-urban inequalities.
As a result, policy responses are often reactive rather than proactive, limiting their effectiveness in mitigating risks associated with informality, such as exposure to environmental hazards or exclusion from essential services (Hussainzad & Gou, 2024)

New technologies, including high-resolution remote sensing data have enabled researchers to map informal areas using texture, density, and morphological indicators, allowing for large-scale assessments across cities and regions (Graesser et al., 2012). Yet, these approaches are often limited in their ability to capture the street-level reality of informal environments—facade conditions, building materials, signage, or human activity—elements that are critical for understanding the lived experience of informality (Cinnamon, 2024). Furthermore, satellite imagery provides only a planar view of urban form, making it difficult to assess vertical growth, structural quality, or gradual processes of upgrading and consolidation. Temporal analyses, while possible through multi-date imagery, often lack the granularity needed to detect subtle but meaningful transformations occurring at the scale of individual streets or dwellings.

Recent advances in computer vision and the availability of large-scale, time-stamped imagery from platforms like Google Street View (GSV) offer a new opportunity to complement traditional remote sensing approaches. Street-level imagery provides a ground-based perspective that reveals details inaccessible from above, such as construction materials, the presence of public infrastructure, vegetation, or social activity (Li et al., 2022). Combined with deep learning models, these images can be automatically analyzed to detect changes in built environments over time, enabling the systematic measurement of upgrading processes, infrastructure provision, and environmental improvements in informal settlements.

Nevertheless, critical methodological questions emerge regarding the capacity of street-level imagery to reliably measure informal settlements. First, concerning spatial and temporal coverage: to what extent does Google Street View (GSV) imagery capture informal settlements? Given that photographs are primarily taken from Google vehicles, do these platforms systematically avoid such areas, or do they provide only peripheral coverage? While GSV data have been successfully used to construct temporal visual datasets documenting neighborhood evolution in formal urban contexts (Li, 2021), the feasibility and temporal depth of such analyses in informal settlements remain uncertain. What is the temporal representation of these areas, and what are the inherent limitations of this data source in capturing their dynamic nature?

Second, street-view imagery as a source of urban change indicators presents well-documented methodological challenges (Hou & Biljecki, 2022). These challenges may be particularly acute in informal settlements, which undergo continuous transformation of their built environment. Understanding what can be reliably extracted from these images to measure change—and under what conditions—is essential for developing robust indicators of upgrading, consolidation, or decline.

This research project aims to address these questions through a case study of Argentina, where national registry data from ReNaBaP (Registro Nacional de Barrios Populares) provide verified geospatial boundaries of informal settlements. By integrating GSV imagery with machine learning techniques, this project develops a methodology to quantify the evolution of the built environment in these areas. Our results are aimed to allow indicators that will be useful to analyze the impacts of both policy interventions and organic growth processes.

Proposal Organization

This proposal is structured as follows. We first present the research objectives, outlining the methodological, empirical, and applied contributions of the project. A targeted literature review then situates the study within the emerging field of street-level urban change analysis, highlighting advances in computer vision, temporal imagery analysis, and applications to informal settlement monitoring. The methodology section describes the data sources—Google Street View imagery, ReNaBaP registry data, and OpenStreetMap—and details the analytical workflow, including panorama clustering, semantic segmentation, and change indicator computation. We present pilot study findings from Tres de Febrero, Buenos Aires, demonstrating the feasibility of the approach across 14 informal neighborhoods and documenting spatial and temporal coverage patterns. The proposal concludes by outlining the planned extension to all 6,467 ReNaBaP neighborhoods nationwide and the anticipated research outputs, including a peer-reviewed publication and an open-source codebase. A methodological appendix provides technical details on the computational procedures employed.

Research Objectives

1. Develop and validate a methodology to quantify physical changes in informal settlements using street-level imagery

- Design a reproducible workflow integrating Google Street View imagery, semantic segmentation models, and geospatial analysis
- Establish protocols for temporal alignment of multi-date panoramas through clustering algorithms
- Generate quantitative indicators of built environment change based on pixel-level semantic classification

2. Systematically assess the spatial and temporal coverage of street-level imagery in informal settlements

- Evaluate the extent to which GSV imagery captures informal settlements registered in Argentina's ReNaBaP database
- Quantify coverage limitations through street density and image coverage indicators
- Identify patterns of data availability and gaps across different settlement typologies and geographic regions

3. Characterize the capabilities and constraints of street-level imagery for monitoring urban transformation in informal contexts
- Document methodological challenges specific to informal settlements, including irregular street networks, variable image quality, and temporal sparsity
- Assess the reliability of visual change indicators under different coverage scenarios
- Establish guidelines for interpreting results and validating detected changes

4. Implement and scale the methodology across all ReNaBaP neighborhoods in Argentina
- Conduct a pilot study in Tres de Febrero to validate the approach and refine analytical procedures
- Extend the analysis to all 6,467 registered informal settlements nationwide
- Generate a comprehensive longitudinal dataset documenting street-level evolution at the settlement level

5. Produce open-access outputs that enable reproducibility and broader application
- Publish peer-reviewed research documenting the methodology, findings, and policy implications
- Release an open-source codebase with complete workflow implementation
- Create a framework adaptable to other urban contexts, cities, or countries with similar data availability

Literature Review on Street Level Change Analysis

Urban monitoring has long relied on traditional data sources such as censuses, household surveys, and administrative records. While these instruments provide valuable demographic and socioeconomic information, they are typically infrequent, costly, and spatially coarse (Khoun et al., 2025). With the acceleration of urbanization, especially in low- and middle-income countries, such approaches have proven insufficient to capture the rapid transformations that cities experience at the neighborhood scale.

Remote sensing emerged as a solution to these limitations by enabling the systematic observation of urban growth from space. High-resolution satellite imagery and aerial photography have been used to map built-up areas, classify land use, and estimate urban density. These methods have greatly advanced our understanding of large-scale urban expansion and its relationship to demographic change (Wang et al., 2021). Yet, their perspective is inherently top-down: they observe roofs, not façades, and surfaces, not materials or human activity. Consequently, important aspects of urban quality—building conditions, infrastructure presence, public space maintenance, or signs of upgrading—remain largely invisible from above (Yu, 2025).

This gap between overhead observation and ground-level reality has motivated a shift toward multi-view urban monitoring, where ground-based imagery complements satellite data. By incorporating the human perspective, researchers can capture the experiential and morphological dimensions of urban space. Street-level imagery offers the potential to observe how people actually inhabit and modify their environments, providing a richer and more nuanced picture of urban change (Cinnamon & Jahiu, 2021).

The development of large-scale platforms such as Google Street View (GSV) has opened unprecedented opportunities for spatial analysis. Since its launch, GSV has amassed billions of geo-referenced and time-stamped panoramas covering cities across the world. The consistent spatial alignment and global reach of this imagery make it a valuable, though underused, dataset for studying urban transformation (Cinnamon & Jahiu, 2021).

Researchers have used GSV for a wide variety of applications: estimating urban greenery and environmental exposure (Lu, 2019), assessing walkability (Kang et al., 2023), evaluating perceived safety and aesthetic appeal, and measuring indicators of socioeconomic status (Fan et al., 2023). In these studies, the imagery acts as a proxy for local environmental conditions—capturing building façades, sidewalks, signage, and vegetation with a level of detail that complements satellite observations.

Crucially, GSV provides temporal coverage, allowing comparisons of the same street scenes across multiple years. This longitudinal dimension transforms street-level imagery from a static visualization tool into a dynamic data source for measuring change. The “time machine” feature of GSV enables researchers to reconstruct urban evolution through time, observing processes such as construction, renovation, vegetation growth, or decay (Ali-bey et al., 2022). Despite differences in camera alignment, weather, and lighting, the potential for systematic temporal analysis of street-level imagery has inspired new research on visual urban dynamics.

The detection of change in visual data is a well-established field in remote sensing and computer vision. Early approaches to change detection relied on spectral or radiometric differences between images, such as image subtraction, principal component analysis, or post-classification comparison. While effective for satellite imagery, these methods are not directly transferable to street-level scenes, where variation in lighting, occlusion, and viewpoint can dominate the signal (You et al., 2020).

Recent progress in deep learning has transformed how visual change is quantified. Semantic segmentation models—such as U-Net, DeepLab, or Mask R-CNN—enable the decomposition of an image into meaningful classes (buildings, vegetation, sky, vehicles, etc.). When applied to multiple temporal instances, these segmentations allow the computation of category-specific change metrics, such as the proportion of built area visible in the frame or the ratio of vegetation to impervious surface (Ulku & Akagündüz, 2022). Such metrics provide interpretable indicators of environmental and structural transformation.

More advanced methods use Siamese networks and temporal contrastive learning to compare images directly. These architectures learn representations that emphasize structural differences rather than superficial visual changes, allowing robust detection of construction, demolition, or façade alterations (Huang et al., 2024).

As multi-temporal street-level imagery has become more accessible, several frameworks for urban change detection have emerged. Early efforts focused on reconstructing city-scale visual evolution by aligning panoramas across time and detecting differences in color or texture patterns (Hu et al., 2025). Later studies introduced semantic-based change detection, where segmentation masks were compared across years to identify alterations in the built environment, such as new construction, façade renovations, or changes in land use visible from the street (Huang et al., 2024).

These frameworks generally follow a common pipeline: (1) acquire multi-date GSV panoramas, (2) ensure geometric consistency by matching viewpoint and heading, (3) apply semantic segmentation or feature extraction, and (4) compute change metrics either at the pixel or object level. Aggregating such changes across space enables the construction of urban change maps that capture transformation processes invisible to overhead sensors.

Nevertheless, this approach faces methodological challenges. Differences in camera position and heading between GSV captures can introduce apparent changes that are not real. Moreover, variations in lighting, occlusion (e.g., passing vehicles or trees), and seasonal conditions complicate the interpretation of visual differences (Hou & Biljecki, 2022). Addressing these issues requires geometric normalization, façade orientation adjustment, and often manual or algorithmic filtering of low-quality images. Recent innovations include the use of building footprints or street network data to align panoramas with the corresponding façades, ensuring that comparisons are made from consistent viewpoints (Li et al., 2025; Roussel et al., 2024).

While most existing work has focused on formal urban environments, the potential of street-level imagery to study informal settlements is increasingly recognized. Informal areas are characterized by irregular layouts, self-built housing, and rapid physical change—conditions that challenge both remote sensing and traditional survey methods. From a street perspective, informal settlements display distinctive visual signatures: heterogeneous building materials, varying façade quality, unpaved streets, and limited public infrastructure (Najmi et al., 2022).

By applying semantic segmentation and visual change detection to street-level imagery, researchers can begin to quantify these characteristics over time. For example, increases in the proportion of masonry walls or reductions in exposed soil could signal upgrading processes. The presence of new streetlights, sidewalks, or vegetation could indicate improved infrastructure or environmental conditions. Such indicators move beyond binary classifications of “formal” versus “informal” and allow for continuous measurement of progress along the upgrading trajectory.

Empirical studies have demonstrated that GSV imagery can capture meaningful indicators of living conditions and infrastructure provision, even in low-income areas (Fan et al., 2023). Yet, few have systematically analyzed temporal sequences to measure change. Integrating street-level change detection into the monitoring of informal settlements would thus fill a major gap between qualitative field observation and coarse-scale remote sensing analyses. It offers the possibility of assessing upgrading interventions, evaluating the pace of consolidation, and detecting early signs of decline or displacement.

The convergence of remote sensing, street-level imagery, and machine learning is opening a new frontier in urban analytics. Each source contributes complementary perspectives: satellites provide synoptic coverage; aerial imagery offers intermediate-scale detail; and street-level data reveal human-scale transformations. Integrating these layers enables a multi-scale, multi-temporal understanding of urban change that links neighborhood processes with citywide trends (Akiner et al., 2025).

Within this landscape, street-level change analysis represents a key methodological innovation. It translates computer vision outputs into interpretable indicators of the built environment, bridging the gap between data-driven analytics and policy-relevant evidence (Zhong et al., 2025). For informal settlements, where data scarcity is particularly acute, the ability to monitor change continuously and non-invasively holds significant promise. It can support local authorities in prioritizing upgrading programs, assist international organizations in tracking progress toward sustainable development goals, and empower communities by making visible the transformations that official statistics often overlook.

However, challenges remain. Street-level data availability is uneven across the Global South, and imagery may not always align temporally with periods of rapid change. Ethical considerations—such as privacy, representation, and consent—must also be addressed. Furthermore, algorithms trained on formal urban contexts may underperform in informal settings, where visual features differ markedly. Future research should therefore focus on developing context-sensitive models, leveraging open data such as Overture or OpenStreetMap to refine façade orientation and block-level analysis, and combining visual indicators with socioeconomic or environmental datasets for a more holistic assessment of urban change.

Data and Methodology

Data

This study integrates multiple spatial data sources to analyze street-level environmental change within informal settlements in Argentina. The three primary datasets used are: (1) Google Street View (GSV) imagery, which serves as the main input for computer vision models; (2) ReNaBaP polygons, which delineate the spatial extent of informal settlements across the country; and (3) OpenStreetMap (OSM) data, which provide a detailed representation of the street network and urban geometry. Together, these datasets enable a multi-scalar analysis that connects visual street-level information with spatial delineations of informality and infrastructural context.

Google Street View Imagery

Google Street View (GSV) provides panoramic imagery captured at street level by vehicles equipped with multi-directional cameras. Each panorama is associated with geographic coordinates (latitude and longitude), a timestamp corresponding to the capture date, and metadata describing camera orientation and other acquisition parameters. In this study, GSV imagery constitutes the foundational dataset for detecting and quantifying changes in the built environment through time.

To construct the dataset, panorama points were programmatically retrieved using the Google Street View API, focusing on panoramas around informal settlements as defined by the ReNaBaP polygons. Each panorama was downloaded for all available years, enabling a longitudinal view of the same location across multiple time periods. The temporal dimension of GSV imagery allows for the observation of neighborhood transformations such as building consolidation, facade improvements, and infrastructure upgrades.

To ensure spatial precision and consistency, all panoramas that are close together are assigned to a cluster.. This correction mitigates potential inaccuracies due to small GPS offsets in GSV metadata. Each corrected panorama point thus lies on a street centerline, ensuring its spatial relationship to surrounding built structures can be accurately modeled. The study further leverages the heading (azimuth) and field of view (FOV) parameters available from GSV metadata to reconstruct the perspective direction of each image. These parameters are crucial for analyzing facade-facing imagery and for computing consistent visual comparisons across time.

ReNaBaP: Registro Nacional de Barrios Populares

The Registro Nacional de Barrios Populares (ReNaBaP) is a national dataset established by the Government of Argentina to identify, delineate, and characterize informal settlements across the country. It includes spatial polygons representing “barrios populares”, along with descriptive attributes such as settlement name, province, locality, and administrative identifiers. The ReNaBaP was designed to support the *Programa de Integración Socio-Urbana*, providing the most comprehensive and official spatial inventory of informal settlements in Argentina.

For the purposes of this study, ReNaBaP polygons are used as the primary spatial units of analysis. Each polygon represents an area with recognized characteristics of informality—such as irregular land tenure, limited infrastructure, and noncompliance with urban planning regulations. By overlaying GSV panorama points onto these polygons, we identify which images fall within or in the immediate vicinity of informal settlements. This spatial linkage allows for systematic analysis of street-level changes in environments that are specifically relevant to the study’s focus on upgrading and consolidation processes.

Furthermore, ReNaBaP polygons are used to compute summary statistics at the settlement level. These include measures of GSV coverage (i.e., the proportion of street segments within the settlement boundary that have Street View imagery available) and spatial heterogeneity in visual features. This enables comparisons between settlements with varying levels of visibility and documentation, highlighting potential biases in data availability.

OpenStreetMap

OpenStreetMap (OSM) provides a globally maintained, open-access vector database of geographic features, including streets, buildings, and land use categories. In this study, OSM data play an auxiliary but essential role. The street network from OSM is used to perform multiple operations that underpin both the spatial alignment of GSV data and the calculation of coverage and visibility metrics.

First, the OSM street geometries serve as a spatial scaffold to pinpoint and snap GSV panorama locations. This ensures that imagery is correctly associated with street centerlines and helps in deriving consistent distances and bearings for computing headings toward built-up areas. Second, the OSM network allows for the estimation of Street View coverage, defined as the ratio of street length with available GSV panoramas to the total street length within each ReNaBaP polygon. This indicator provides an estimate of how well each informal settlement is represented in the GSV database, revealing spatial inequalities in digital visibility.

Additionally, OSM data are used to assist in generating building-facing headings and fields of view (FOV). By clipping the street network to each ReNaBaP polygon, it is possible to estimate the most relevant orientations for image capture—favoring building facades over open roads. This process allows the methodology to adapt to complex street geometries, including narrow passages or irregular block shapes typical of informal settlements.

Finally, OSM building footprints and other geometries can be used for validation and contextualization, supporting cross-referencing with the visual information extracted from Street View imagery. This helps ensure that detected changes correspond to genuine transformations in the built environment rather than artifacts of image orientation or capture inconsistencies.

Image Coverage Indicators

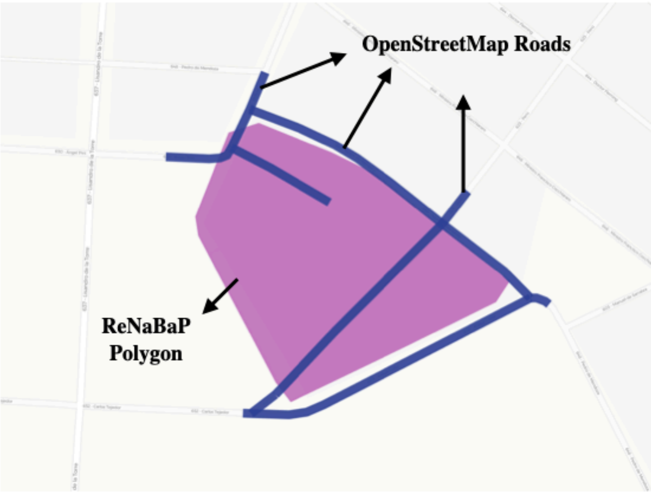
A critical prerequisite for using street-level imagery to monitor informal settlements is understanding the spatial extent and quality of available data. Coverage limitations may arise from two distinct sources: first, the absence or limited density of street infrastructure within informal settlements, which constrains the accessibility of imaging vehicles; and second, the incomplete or sparse capture of existing streets by imaging platforms. To systematically assess these dimensions of data availability, we propose two complementary indicators.

1. Street Density Indicator

This indicator quantifies the availability of street infrastructure within each informal settlement. It is calculated as the ratio of total street length (in meters) to the settlement's surface area (in square meters), expressed as:

Street Density = Total Street Length (m) / Settlement Area (m²)

Higher values indicate more developed street networks, which typically provide greater accessibility for both imaging vehicles and analytical coverage. Conversely, low street density may reflect limited formal infrastructure, constraining the potential for comprehensive street-level monitoring. This metric is computed using street network data from OpenStreetMap or official cadastral sources, intersected with ReNaBaP settlement boundaries.



Component	Value
Meters of Street	692.7
Area of Polygon (m²)	24,588.6
Street Density *(meters / m²)*	0.028

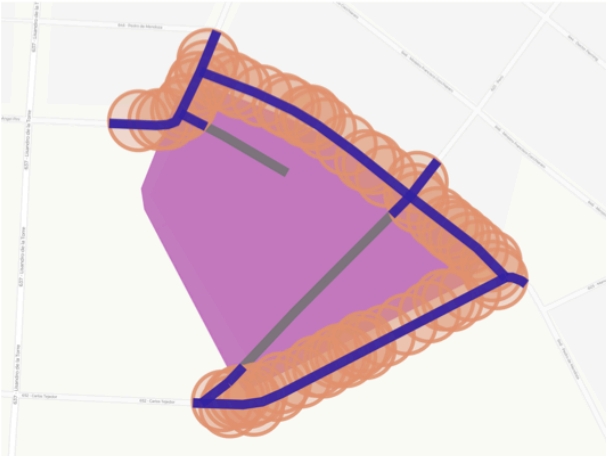
Figure: Street density indicator computation, calculated as the length of streets divided by total area of the neighborhood. Case study ReNaBaP 313

2. Image Coverage Indicator

This indicator measures the proportion of the existing street network that is captured by available GSV panoramas. To calculate it, we identify all georeferenced panorama locations within each settlement and generate a 10-meter observation buffer around each point, representing the approximate effective viewing distance of a street-level image. The total length of streets falling within these buffers is then divided by the total street length within the settlement:

Image Coverage = Street Length Captured by Images (km) / Total Street Length (km)

Values approaching 1.0 indicate near-complete visual coverage of the street network, while lower values signal gaps in imagery that may limit the feasibility of temporal change analysis. This metric allows us to distinguish between settlements that lack street infrastructure and those that have streets but insufficient imagery, informing the applicability and reliability of subsequent visual analytics.



Component	Value
Meters of Street Observable in GSV (*blue*)	537.8
Total Meters of Street (*blue + gray*)	692.7
Coverage	0.776

Figure: Image coverage Indicator. ReNaBaP 313, Tres de Febrero, Buenos Aires

The figure illustrates the spatial distribution of available GSV panoramas and their corresponding 10-meter observation buffers, demonstrating the extent to which the existing street network is captured by imagery in representative informal settlements.

Built-up Segmentation and Growth indicators

Our methodology of built-up growth from street level imagery involves a series of procedures including: (1) the selection and clustering of street level imagery (panoramas); (2) the heading and field-of-view (FOV) positioning to ensure the comparability of images; (3) the computer vision–based feature extraction methods; and (4) the change detection indicators definitions.

For the sake of brevity we discuss the details in the Appendix. We focus here on describing the results that can be obtained from the procedure.

Once the GSV images are obtained, the next step involves extracting semantic information from each image to characterize its visual composition. For this purpose, the STEGO (Self-Supervised Transformer with Energy-Based Graph Optimization) model is employed. STEGO is a self-supervised segmentation model capable of producing pixel-level classifications by learning semantic representations from large datasets without requiring extensive labeled data.

Figure: 4198 Pedro de Mendoza street, Caseros, Buenos Aires, December 2013

Figure: 4198 Pedro de Mendoza street, Caseros, Buenos Aires, May 2024

In this study, STEGO is applied using its pretrained configuration on the COCO-Stuff dataset, which includes a broad set of urban-related classes such as buildings, vegetation, sky, road, and vehicles. The model segments each image into multiple semantic classes, and the output is processed to compute the proportion of pixels belonging to each class (e.g., buildings = 15%, sky = 30%, trees = 12%, etc.).

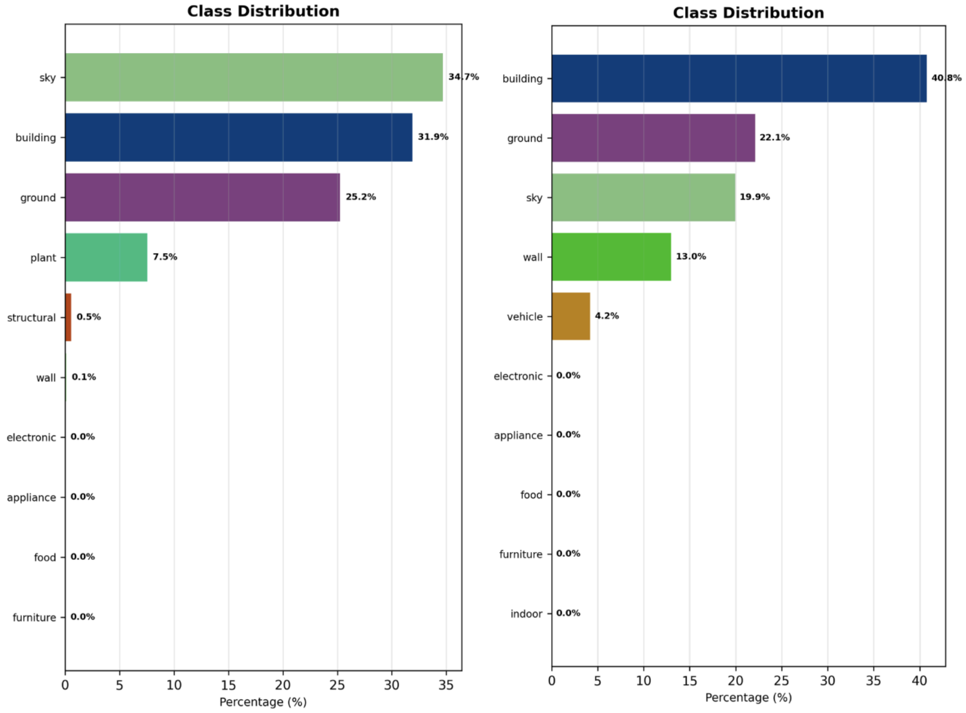


Figure: Segmented class distributions for 4198 Pedro de Mendoza street, Caseros, Buenos Aires. Year 2013 (left) shows 31.9% of pixels belonging to the "building" class, while year 2024 (right) shows 40.8%, almost a 10% increase.

To improve the robustness of class predictions and account for minor differences in lighting or orientation, a linear calibration function is applied to the model's softmax outputs, aligning the confidence scores across temporal instances. This produces a stable, interpretable representation of the visual environment at each viewpoint, enabling comparisons across time in a compact numerical form.

[Explain issues to solve]
The need to correctly point to facades before calculating % of pixels for each class

Pilot Study Findings

The pilot study was conducted in the Tres de Febrero department, located in the Buenos Aires Province, Argentina. This area was selected because it contains a diverse urban fabric that includes both consolidated residential zones and informal settlements registered in the ReNaBaP (Registro Nacional de Barrios Populares) database. Within Tres de Febrero, a total of 14 ReNaBaP neighborhoods were identified and used as the core spatial units of analysis.

To ensure that contextual surroundings were also captured—such as bordering streets, adjacent land uses, and visual transitions between formal and informal areas—the area of interest was defined as the union of all neighborhood polygons plus a 500-meter buffer. Within this extended area, the GSV API returned over 85,000 unique panorama points, which represent the total set of available street-level images over time. These panoramas were then filtered to identify those located inside or near the neighborhood polygons, including those from which the neighborhood area was visible (even if the camera was positioned slightly outside the official boundary). After filtering, a total of 848 relevant panoramas were retained for subsequent analysis.

ID	Name	Locality	Approx. House-holds	Approx. Families	Surface (m^2)	Number of GSV Panos
299	La Favela	El Libertador	40	44	5,571	16
313	Mercado	Caseros	130	143	10,233	97
398	Villa Risso	Caseros	14	15	2,394	50
1025	Once de Septiem-bre	Churruca	30	33	6,910	36
1026	Rayito de Sol	Remedios De Escalada	165	182	19,615	96
1028	Puerta 8	Churruca	170	187	15,318	27
1087	París	Loma Hermosa	50	55	5,349	28
1107	Capitán Bermúdez	Caseros	150	165	5,133	30
1122	Paredón	Ciudadela	100	110	13,656	114
2929	Villa Matheus	Caseros	80	88	4,604	24
4297	Maldonado	Ciudadela	600	660	12,854	207
4614	Derqui	Caseros	25	28	3,109	63
5385	El Progreso	Ciudadela	12	13	4,589	24
5386	Cuba	Loma Hermosa	15	17	1,522	36

Figure: ReNaBaP neighborhoods in Tres de Febrero department, Buenos Aires

The spatial distribution of these panoramas is uneven: most neighborhoods contain fewer than 50 panoramas, which corresponds to smaller or less accessible areas, whereas a few neighborhoods contain more than 90 panoramas, typically those with denser street networks, higher GSV coverage, or multiple capture campaigns over time.

Coverage

In terms of coverage, defined as the proportion of total street length within or near the neighborhood that is covered by GSV imagery, results indicate that most neighborhoods have coverage above 60%, suggesting that the majority of street segments within the study area are represented in the imagery. Among the 14 neighborhoods, 9 exhibit full (100%) coverage, meaning that all accessible streets have at least one GSV panorama, while 3 neighborhoods have no coverage at all (0%). These cases of missing coverage can be attributed to several factors. Some neighborhoods have limited or informal street layouts that are not accessible to GSV vehicles, which typically require motorable paths. In other cases, street visibility restrictions—due to private roads, gated sectors, or ongoing urban works—may prevent imagery collection. Another possibility is temporal mismatch, where neighborhoods were only recently incorporated into the urban fabric or improved, and GSV has not yet updated its coverage in those zones.

ReNaBaP ID	Coverage	Street Density (m/m^2)
299	0.475	0.024370
313	0.776	0.028175
398	1.000	0.031709
1025	0.608	0.044610
1026	0.671	0.020279
1028	0.623	0.027168
1033	0.000	0.030979
1087	1.000	0.024431
1090	1.000	0.004328
1107	0.768	0.023555
1122	1.000	0.021919
2878	0.602	0.026126
2929	1.000	0.018023
4297	1.000	0.029436
4613	0.000	0.010023
4614	1.000	0.031809
5384	0.000	0.000000
5385	1.000	0.014440
5386	1.000	0.027358
5387	0.426	0.014664

Figure: Coverage and Street Density metrics by ReNaBaP neighborhoods

Time coverage

After applying the DBSCAN clustering algorithm to the filtered panoramas, each cluster was expected to represent a unique physical location captured at different points in time. The results confirm this assumption: nearly all clusters include at least two panoramas taken at distinct dates, often spanning multiple years. This temporal variation is critical, as it enables the identification of infrastructure changes through time-lapse analysis.

The clustering step effectively groups panoramas that are within approximately 2.5 meters of each other—accounting for small GPS drifts or camera repositioning between capture campaigns—while ensuring that only clusters with at least two observations are retained for temporal analysis. As a result, almost 100% of the clusters exhibit valid temporal coverage, providing a solid foundation for studying changes in the built environment.

The temporal structure of the data also varies across neighborhoods: areas with higher accessibility or greater importance in the urban network tend to have more frequent GSV revisits, resulting in clusters with up to 10 or more temporal snapshots. Conversely, smaller or less accessible neighborhoods may have only two or three temporal observations, which still allows for basic change detection but limits the granularity of temporal analysis.

Planned extension to Argentina's neighborhoods

Building on the results of the pilot study conducted in Tres de Febrero, the next objective is to scale the methodology to all ReNaBaP neighborhoods across Argentina. This expansion seeks to systematically assess the spatial and temporal dynamics of informal settlements nationwide, leveraging Google Street View imagery and computer vision models to generate consistent, scalable indicators of physical change. By applying the same methodological framework—panorama clustering, heading generation, semantic segmentation, and change detection—across thousands of neighborhoods, the project aims to construct a longitudinal dataset that captures the evolution of informality at the street level. Such a dataset would serve as a valuable tool for policymakers, researchers, and civil society organizations working to monitor urban upgrading, infrastructure provision, and environmental conditions in vulnerable communities.

Province	Number of Neighborhoods
Buenos Aires	2065
Catamarca	61
Chaco	442
Chubut	83
Ciudad Autónoma de Buenos Aires	49
Corrientes	257
Córdoba	318
Entre Ríos	231
Formosa	126
Jujuy	159
La Pampa	9
La Rioja	34
Mendoza	360
Misiones	413
Neuquén	116
Río Negro	240
Salta	354
San Juan	81
San Luis	32
Santa Cruz	21
Santa Fe	469
Santiago del Estero	128
Tierra del Fuego	49
Tucumán	370

Figure: ReNaBaP Neighborhoods in Argentinian Provinces.

Argentina's Registro Nacional de Barrios Populares (ReNaBaP) currently identifies 6,467 informal neighborhoods distributed across all 24 provinces of the country. The province of Buenos Aires accounts for the largest share, with approximately 2,065 mapped neighborhoods, followed by other urbanized provinces such as Chaco, Salta, Tucumán, Córdoba, and Misiones. Extending the method to this national scale presents both an analytical and operational challenge, given the heterogeneity in data availability, street network structure, and imagery coverage across regions. Nevertheless, achieving this level of coverage would enable the first nationwide, street-level characterization of informal settlements, bridging the gap between traditional remote sensing and on-the-ground monitoring. The resulting dataset would not only provide new insights into spatial inequality and urban transformation but also support evidence-based decision-making aligned with national housing and territorial development agendas.

Research Outputs

The project will generate two primary research outputs: a peer-reviewed research paper and an open-source codebase. The paper will document the methodological framework, pilot implementation, and key findings, contributing to the growing literature on street-level urban monitoring and informal settlement analysis. In parallel, the complete workflow—from data acquisition and clustering to segmentation and change detection—will be made publicly available as an open-source repository. This transparency not only promotes reproducibility but also enables other researchers and practitioners to adapt and extend the methodology to different contexts, cities, or countries.

References

Acevedo-De-los-Ríos, A., Chumpitaz-Requena, F. R., Rondinel-Oviedo, D. R., Cárdenas-Mamani, Ú., & Redondo, J. M. (2025). Exploring the urban systemic scenarios of improving socioecological conditions in an informal settlement of a developing country with a system dynamics model. Environmental Science & Policy, 169, 104079.

Akiner, M. E., & Ghasri, M. (2025). Integrating Remote Sensing and Machine Learning to Analyze Urban Growth and Its Environmental Effects: A 30-Year Assessment in Başakşehir, Turkey. Pure and Applied Geophysics, 1-25.

Ali-bey, A., Chaib-draa, B., & Giguere, P. (2022). Gsv-cities: Toward appropriate supervised visual place recognition. Neurocomputing, 513, 194-203.

Cinnamon, J. (2024). Visual imagery and the informal city: examining 360-degree imaging technologies for informal settlement representation. Information Technology for Development, 30(4), 590-607.

Cinnamon, J., & Jahiu, L. (2021). Panoramic Street-Level Imagery in Data-Driven Urban Research: A Comprehensive Global Review of Applications, Techniques, and Practical Considerations. ISPRS International Journal of Geo-Information, 10(7), 471. https://doi.org/10.3390/ijgi10070471

Fan, Z., Zhang, F., Loo, B. P., & Ratti, C. (2023). Urban visual intelligence: Uncovering hidden city profiles with street view images. Proceedings of the National Academy of Sciences, 120(27), e2220417120.

Graesser, J., Cheriyadat, A., Vatsavai, R. R., Chandola, V., Long, J., & Bright, E. (2012). Image based characterization of formal and informal neighborhoods in an urban landscape. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 5(4), 1164-1176.

Habitat, U. N. (2016). Leveraging land: Land-based finance for local governments a reader.

Hou, Y., & Biljecki, F. (2022). A comprehensive framework for evaluating the quality of street view imagery. International Journal of Applied Earth Observation and Geoinformation, 115, 103094.

Hu, A., Yabuki, N., & Fukuda, T. (2025). Multi-temporal analysis of urban vegetation using deep learning and 3D reconstruction. Landscape Ecology, 40(7), 125.

Huang, T., Wu, Z., Wu, J., Hwang, J., & Rajagopal, R. (2024, March). Citypulse: Fine-grained assessment of urban change with street view time series. In Proceedings of the AAAI Conference on Artificial Intelligence (Vol. 38, No. 20, pp. 22123-22131).

Huchzermeyer, M., & Karam, A. (Eds.). (2006). Informal settlements: A perpetual challenge?. Juta and Company Ltd.

Hussainzad, E. A., & Gou, Z. (2024). Climate risk and vulnerability assessment in informal settlements of the global south: a critical review. Land, 13(9), 1357.

Kang, Y., Kim, J., Park, J., & Lee, J. (2023). Assessment of perceived and physical walkability using street view images and deep learning technology. ISPRS International Journal of Geo-Information, 12(5), 186.

Khoun, T., Poortinga, A., Thwal, N. S., De Alba, I. G., McMahon, A., & Mendez, C. (2025). Mapping the dimensions of poverty through big data, socioeconomic surveys and machine learning in Cambodia. Social Indicators Research. https://doi.org/10.1007/s11205-025-03718-3

Li, W., Yu, J., Chen, D., Lin, Y., Dong, R., Zhang, X., ... & Fu, H. (2025). Fine-grained building function recognition with street-view images and GIS map data via geometry-aware semi-supervised learning. International Journal of Applied Earth Observation and Geoinformation, 137, 104386.

Li, X. (2021). Examining the spatial distribution and temporal change of the green view index in New York City using Google Street View images and deep learning. Environment and Planning B: Urban Analytics and City Science, 48(7), 2039-2054.

Li, Y., Peng, L., Wu, C., & Zhang, J. (2022). Street View Imagery (SVI) in the built environment: A theoretical and systematic review. Buildings, 12(8), 1167.

Lu, Y. (2019). Using Google Street View to investigate the association between street greenery and physical activity. Landscape and Urban Planning, 191, 103435.

Mahendra, A., King, R., Du, J., Dasgupta, A., Beard, V. A., Kallergis, A., & Schalch, K. (2021). Seven transformations for more equitable and sustainable cities. World resources report, towards a more equal city.

Medellín's Integrated Urban Projects - Lincoln Institute of Land Policy. (2024, May 29). Lincoln Institute of Land Policy. https://www.lincolninst.edu/publications/working-papers/medellins-integrated-urban-projects/

Mwanza, C. (2024). State of Efforts to Realize Adequate Housing for All: A UN-Habitat Report.

Najmi, A., Gevaert, C. M., Kohli, D., Kuffer, M., & Pratomo, J. (2022). Integrating remote sensing and street view imagery for mapping slums. ISPRS International Journal of Geo-Information, 11(12), 631.

Public land ownership and leasing in Helsinki, Finland - Housing2030. (2021, October 4). Housing2030. https://www.housing2030.org/project/public-land-ownership-and-leasing-in-helsinki-finland/

Roussel, R., Jacoby, S., & Asadipour, A. (2024). Robust building identification from street views using deep convolutional neural networks. Buildings, 14(3), 578.

Sartori, G., Nembrini, G., & Stauffer, F. (2002). Monitoring of urban growth of informal settlements and population estimation from aerial photography and satellite imaging. Occasional paper, 6.

Smit, W. (2021). Urbanization in the global south. In Oxford research encyclopedia of global public health.

Thomson, D. R., Kuffer, M., Boo, G., Hati, B., Grippa, T., Elsey, H., ... & Kabaria, C. (2020). Need for an integrated deprived area "slum" mapping system (IDEAMAPS) in low-and middle-income countries (LMICs). Social Sciences, 9(5), 80.

Ulku, I., & Akagündüz, E. (2022). A survey on deep learning-based architectures for semantic segmentation on 2d images. Applied Artificial Intelligence, 36(1), 2032924.

United Nations Department of Economic and Social Affairs (2025). The Sustainable Development Goals Report 2025. New York.

Wang, H., Gong, X., Wang, B., Deng, C., & Cao, Q. (2021). Urban development analysis using built-up area maps based on multiple high-resolution satellite data. International Journal of Applied Earth Observation and Geoinformation, 103, 102500.

You, Y., Cao, J., & Zhou, W. (2020). A survey of change detection methods based on remote sensing images for multi-source and multi-objective scenarios. Remote Sensing, 12(15), 2460.

Yu, D. (2025). Toward Integrated Urban Observatories: Synthesizing Remote and Social Sensing in Urban Science. Remote Sensing, 17(12), 2041. https://doi.org/10.3390/rs17122041

Zhong, W., Wang, L., Han, X., & Gao, Z. (2025). Spatiotemporal Analysis of Urban Perception Using Multi-Year Street View Images and Deep Learning. ISPRS International Journal of Geo-Information, 14(10), 390.

Appendix - Methodology

This section describes the methodological workflow developed to analyze street-level changes within informal settlements using Google Street View (GSV) imagery. The process involves four main steps: (1) panorama selection and clustering; (2) heading and field-of-view (FOV) generation; (3) computer vision–based feature extraction; and (4) change detection. Together, these steps enable the spatial alignment, visual interpretation, and temporal comparison of GSV imagery in and around informal settlements identified by the ReNaBaP dataset.

Panorama Selection and Clustering

After retrieving all available GSV panorama points for an area encompassing the ReNaBaP polygons, the dataset typically contains multiple panoramas per location, each representing a different capture date. However, GSV panoramas are not perfectly aligned across time due to small positional drifts caused by changes in camera placement or vehicle trajectory. To ensure that panoramas corresponding to the same physical location are analyzed together, the points are grouped into clusters using the DBSCAN (Density-Based Spatial Clustering of Applications with Noise) algorithm.

DBSCAN identifies clusters as regions of high point density separated by areas of lower density. Unlike partition-based methods such as *k*-means, DBSCAN does not require specifying the number of clusters in advance and can effectively identify clusters of arbitrary shape while discarding noise points. The algorithm uses two parameters: ε (epsilon), which defines the maximum distance between two points to be considered neighbors, and min_samples, which determines the minimum number of points required to form a cluster.

The parameters were selected through manual validation using the GSV web platform to visually assess cluster compactness and temporal consistency. Based on this validation, the parameters $\varepsilon = 2.5$ meters and min_samples = 2 were chosen. These values ensure that panoramas from different years but corresponding to the same viewpoint are correctly grouped, given that GSV capture locations for the same site typically drift only a few meters between campaigns. The minimum of two samples ensures that only locations with at least two distinct temporal observations are retained for longitudinal analysis.

Alternative configurations were also tested with ε = [5, 2.5, 1] and min_samples = [2, 3, 4], but the selected parameters provided the most consistent temporal alignment and minimized over-clustering in dense areas. The resulting clusters closely replicate the behavior of the Street View platform: clusters have between 2 and 16 panorama points, with a mean of 4.26 and a standard deviation of 2.31, corresponding to approximately four capture dates in residential areas and up to ten or more along major roadways and highways.

Each cluster thus represents a single street-view location with multiple temporal observations, forming the foundation for view-level and cluster-level temporal comparisons in subsequent stages.

Heading and Field-of-View (FOV) Generation

In progress

Image Segmentation using STEGO

More details on the algorithm TBC